

# Multi-Agent Financial Research System

## UI/UX Design Specification — Volume 3 of 4

AI Chat · Chat History · Citations Viewer · Financial Metrics · Company Comparison · Red Flag Analysis · Agent Activity Monitor

References Volume 1 for all named components, colors, and tokens.

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### Screen 15 — AI Chat

**Purpose:** The product's core interaction — ask a natural-language financial question and receive a grounded, cited answer from the Research Agent.

**Layout:** Two-pane layout within the Workspace Overview shell.

- **Left/main pane (flex):** Vertical chat transcript, newest at bottom, input composer pinned to bottom.
- **Right pane (360px, collapsible):** Citations Viewer panel (Screen 17), contextually populated with the citation(s) referenced by whichever answer is currently focused/hovered.

**Composer:** Multi-line auto-growing textarea, placeholder rotates through example prompts ("Compare TCS and Infosys profitability," "What are the major business risks?"), Send button (also ) , and a small "scope" chip showing which document(s) the question will search (default: all documents in workspace; clickable to narrow to one).

**Components:** User message bubble, AI answer bubble, Citation Pill, "Thinking" state component, No-answer state bubble, Composer, Scope chip, Suggested-question chips (shown only when transcript is empty).

#### Interactions:

- Typing and sending appends a user bubble instantly, then a "Thinking" bubble (Research Agent avatar, pulsing) replaces it with the answer when ready.
- Clicking a Citation Pill inline opens/updates the right-pane Citations Viewer scrolled to that exact page and highlights the source sentence.
- Multi-part questions render as a single answer bubble with internally numbered sub-answers, each with its own citation pill — visually operationalizing "step-by-step reasoning."
- Long-press/right-click on an answer offers "Copy," "Add to Report," and "Flag as incorrect" (feedback loop).

## States:

- **Empty (first visit to a workspace's chat):** centered suggested-question chips + one-line explainer ("Ask anything about the documents in this workspace — I'll always show you the page it came from.").
- **Thinking:** pulsing agent avatar + 3-dot animation, composer disabled while a request is in flight, "Stop generating" ghost button appears after 3s.
- **Answered — grounded:** standard bubble with  $\geq 1$  citation pill.
- **Answered — not found:** muted-tone bubble, distinct background (`Neutral/100` instead of white) explicitly stating the system found no supporting evidence, with a "Try rephrasing" or "Upload a document that covers this" suggestion.
- **Error (agent failure/timeout):** red-tinted system bubble, "Something went wrong generating this answer" + Retry button.

**Animations:** New bubbles slide-up+fade (160ms). Citation pill hover triggers a 120ms highlight pulse on the corresponding right-pane page section, so the eye connects text ↔ source instantly.

**Responsive:** On mobile, the Citations Viewer right pane becomes a bottom sheet triggered by tapping a citation pill (rather than a persistent side pane), preserving full-width chat.

**Design Rationale:** The always-available right-pane citation view (rather than only a modal on click) lets power users (Ananya, the analyst) verify claims without losing their place in the conversation — critical for the "trust through transparency" goal.

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## Screen 16 — Chat History (Drawer)

**Purpose:** Browse and resume past conversations within a workspace.

**Layout:** Right-side drawer (E3 elevation), list of past sessions grouped by date ("Today," "Yesterday," "Last 7 days," "Earlier"), each entry showing the first user question as its title and a message-count badge.

**Components:** Drawer, list rows (title + timestamp + count), search-within-history input at top.

**Interactions:** Clicking a session scrolls the main chat transcript to that point (sessions are not siloed — chat history is one continuous per-workspace log, grouped visually by session for scanability). Search filters rows by keyword match against question text.

**States:** Populated · Empty ("No conversations yet — ask your first question") · Search no-results.

**Design Rationale:** Treating history as one continuous, searchable log (rather than disconnected "chat sessions" like a generic chatbot) matches the mental model of a research notebook, which is how the target personas actually work.

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## Screen 17 — Citations Viewer (Panel + Modal Variants)

**Purpose:** Let the user verify any AI claim against the literal source page — the product's core trust mechanism.

**Layout — Panel variant (docked, used inline with Chat):**

- Header: document name + page number (e.g., "Infosys\_AR\_2025.pdf · Page 42") + "Open full document" link.
- Body: rendered PDF page image with the specific supporting sentence/paragraph highlighted (yellow highlight overlay, semi-transparent).
- Footer: page navigation (prev/next), zoom controls.

**Layout — Modal variant (used from Financial Metrics/Comparison/Red Flags when clicking a source-page badge):** Same content, centered modal, max-width 800px, with a dimmed backdrop.

**Components:** PDF page renderer, highlight overlay, page navigation controls, zoom controls, "Open full document" link (routes to Document Library detail).

**Interactions:** Prev/next page navigation preserves the highlight only on the originating page; other pages render plain. Zoom persists across page navigation within a session.

**States:** Loading (skeleton page silhouette) · Loaded · Page render failed (fallback: "View raw extracted text instead" toggle, showing plain text with the same highlight applied).

**Animations:** Highlight overlay fades in (200ms) after the page image loads, drawing the eye directly to the evidence.

**Responsive:** Modal variant becomes a full-screen sheet on mobile; panel variant becomes a bottom sheet (see Screen 15 responsive note).

**Design Rationale:** This single component is reused everywhere a citation exists (chat, metrics, comparison, red flags, reports) — one canonical "prove it" interaction across the entire product, rather than a bespoke citation UI per feature.

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## Screen 18 — Financial Metrics Dashboard

**Purpose:** Present the Extraction Agent's structured output — the core financial metrics of a

company — in a scannable, source-linked format.

**Layout:** Per-document (or per-company, if multiple documents belong to the same company) dashboard.

- Row 1: Company header (name, fiscal year selector if multi-year data exists, "not found" count badge if applicable).
- Row 2: Metric Card grid (4 columns desktop) — Revenue, Net Profit, EBITDA, EPS, ROE, ROCE, Debt, Cash Flow, Operating Margin, Gross Margin — each card shows value (mono numerals), YoY delta arrow+percentage, and a small source-page badge (click → Citations Viewer modal).
- Row 3: "Trend" section — line/bar charts for Revenue and Profit across available years (if multi-year), using the data-viz palette.
- Row 4: "Extracted Risk Disclosures" — collapsed accordion of narrative risk text pulled by the Extraction Agent, each entry with its source badge.

**Components:** Metric Card, fiscal-year Select, Trend chart (Recharts line/bar), Accordion, source-page badge.

**Interactions:** Clicking any source-page badge opens the Citations Viewer modal. Hovering a Metric Card reveals a small "How this was calculated" tooltip when the metric is derived (e.g., margin = operating income / revenue) rather than directly extracted.

**States:** Populated · Partial (some metrics "Not Found" — shown as a muted dash "—" with tooltip "Not disclosed in this document," never a blank cell) · Loading (skeleton metric cards) · No metrics yet (Extraction Agent still running — shows the Agent Status Chip inline with an estimated-time caption).

**Animations:** Metric Cards count up from 0 to their final value on first load (500ms ease-out) — a small moment of delight without being gimmicky, since these are the product's "hero numbers."

**Responsive:** 4-column grid → 2-column (tablet) → 1-column (mobile); trend charts remain full-width and horizontally scrollable if needed.

**Design Rationale:** Explicitly marking "Not Found" rather than omitting a metric prevents the dangerous UX failure mode of a user assuming a blank space means "zero" — directly supporting the zero-hallucination product guarantee.

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## Screen 19 — Company Comparison

**Purpose:** Side-by-side benchmarking across two or more companies in a workspace — the Comparison Agent's primary output.

### Layout:

- Header: company multi-select (chips, up to 4 companies), "Compare" primary button.
- Body: Comparison Table (sticky first column of metric labels, one column per selected company), each cell showing value + source badge; best-performer cell per row highlighted with a subtle `Success/100` background and a small trophy/check icon; worst-performer with a subtle `Neutral/100` tint (never alarmist red — underperformance isn't automatically a "red flag").
- Below table: a grouped bar chart visualizing the 3–4 headline metrics (Revenue, Net Profit, Debt, ROE) across companies for quick visual scanning.

**Components:** Multi-select chip input, Comparison Table, source badge, grouped bar chart, Empty-selection state.

**Interactions:** Adding/removing a company from the selection live-updates the table and chart (debounced 300ms, with a brief skeleton overlay on the changed columns only, not the whole table). Column reordering via drag on the table header.

**States:** No companies selected (prompt: "Select 2 or more companies to compare") · Comparing (populated) · A company with missing data for a given metric shows "N/A" in that cell rather than breaking the table layout.

**Animations:** New comparison column slides in from the right (240ms) when a company is added.

**Responsive:** Table becomes horizontally scrollable with the metric-label column frozen (sticky) on mobile/tablet — never collapses metrics into an unreadable stacked card format, since side-by-side comparison is the entire point of the screen.

**Design Rationale:** Explicitly avoiding red/green "good/bad" coloring for comparison (reserved only for actual Red Flags) prevents conflating "underperforms peer" with "financial risk," which are different concepts the product must not blur.

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## Screen 20 — Red Flag Analysis

**Purpose:** Surface the Red Flag Agent's automatically detected risk indicators without requiring the user to ask.

### Layout:

- Header: Severity filter chips (All / High / Medium / Low), company/document filter.
- Body: Vertical list of Red Flag Cards, sorted High → Low severity by default. Each card: Severity Badge, flag type label (e.g., "Increasing Debt"), 1–2 sentence AI-generated

explanation, source-page badge, and a small trend sparkline where applicable (e.g., debt over time).

- Empty (no flags detected): a calm, reassuring state — NOT a blank/empty-state illustration reused from elsewhere, but a distinct green-toned confirmation: "No significant red flags detected across 3 documents" with a checkmark, since this is a positive result the user should trust, not an absence-of-content problem.

**Components:** Severity Badge, Red Flag Card, filter chips, source badge, sparkline.

**Interactions:** Clicking a card expands it in place (accordion) to show the full extracted evidence passage before linking to the full Citations Viewer. Filtering by severity animates out non-matching cards (fade+collapse, 160ms).

**States:** Populated (flags exist) · Reassuring-empty (no flags, described above) · Loading (Red Flag Agent still running — inline Agent Status Chip with caption "Scanning for risks...").

**Animations:** High-severity cards on first load have a single, subtle attention pulse on the Severity Badge only (one-time, not looping) to draw the eye without being alarmist/looping-blinking (which would feel like a system error rather than a financial insight).

**Responsive:** Cards remain full-width single column at all breakpoints (this screen is naturally list-based and doesn't need a grid).

**Design Rationale:** Deliberately distinguishing the "reassuring empty state" from generic empty states (Volume 4) matters because "no results" here is good news, and treating it identically to a cold-start empty state would undersell a genuinely useful negative result.

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## Screen 21 — Agent Activity Monitor

**Purpose:** Full transparency into what each of the 6 agents has done, is doing, and produced — the dedicated home for the "confidence in AI activity" goal.

**Layout:** Timeline/log view, reverse-chronological. Each entry: Agent badge (3D claymorphism icon in identity color), agent name, action taken ("Extracted 9 metrics from Infosys\_AR\_2025.pdf"), duration, timestamp, and a "View Output" link that deep-links to the relevant screen (Extraction → Metrics tab, Red Flag → Red Flags tab, etc.).

- Top summary bar: 6 Agent Status Chips in a row (one per agent) showing current state across the whole workspace at a glance — this is the expanded, full version of the Agent Status Rail from Screen 11.
- Filter: by agent, by status (Complete/Failed/Running), by document.

**Components:** Agent Status Chip ×6 (summary bar), Timeline entry (Agent Output Card variant), filter dropdowns.

**Interactions:** Clicking "Retry" on a Failed entry re-triggers that agent for that document, showing a live-updating status chip in place.

**States:** Populated · Empty (before any document processed: "Agent activity will appear here once you upload a document") · A Running entry shows an inline progress indicator specific to that agent (e.g., Extraction Agent entry shows "Extracting... 6/10 metrics found so far").

**Animations:** New log entries prepend with a slide-down+fade (200ms); the summary bar chips pulse briefly when a status changes.

**Responsive:** Summary bar wraps to 2 rows of 3 chips on tablet, 3 rows of 2 on mobile; timeline entries remain single-column full-width.

**Design Rationale:** This screen exists specifically so that no agent's work is ever a "black box" — every single action taken by every agent is logged and inspectable, which is both a trust feature for end users and a debugging feature for the development team (dual-purpose by design, though the end-user and admin views share this same component with different data scoping).

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*End of Volume 3. Continue to Volume 4: Report Generation, Settings, Profile, Notifications, Error States, Responsive Design, Motion, Developer Handoff.*